Regression Assignment:

1.) Identify your problem statement

predict the insurance charges based on input parameters.

2.) Tell basic info about the dataset (Total number of rows, columns)

Total rows: 1338
Total columns: 6

3.) Mention the pre-processing method if you're doing any (like converting string to number – nominal data)

preprocessing for Sex and Smokers
using get_dummies on column sex
drop_first=True

4.) Develop a good model with r2_score. You can use any machine learning algorithm; you can create many models. Finally, you have to come up with final model.

Please scroll down to see the observation run for various model namely:

- (*) Multi Linear
- (*) SVM
- (*)Decision Tree
- (*)Random Forest

5.) All the research values (r2_score of the models) should be documented. (You can make tabulation or screenshot of the results.)

Please scroll down to see the various model observation

6.) Mention your final model, justify why u have chosen the same.

Model	Best R2 Score	Key Configuration	Performance Rank
SVM	**0.878024**	Kernel: **RBF**, C: **20000**	**1st**
Random Forest	**0.875757**	Criterion: **absolute_error**, Features: **Log2**	**2nd**
Decision Tree	**0.747328**	Criterion: **Poisson**, Splitter: **Random**	3rd
Linear Regression	**0.710092**	Standard Multi-Linear	4th

- SVM achieved the highest mathematical score, proving the RBF Kernel is perfect for the non-linear patterns but with C=20000 is considered very "aggressive." as it tries extremely hard to fit every data point, which leads to overfitting.
- Random Forest, by nature, is an ensemble; it averages multiple trees, making it much more stable, The fact that Random Forest performed best with MAX_FEATURES=log2 means the model is successfully filtering out noise(random variation in the data) in our given data sample and achieved 0.875 without needing all the features all at once.
- If our dataset is strictly static and if we only care about the absolute highest precision for this specific set, then the SVM is good.

<u>Conclusion:</u>

Even though ranking vise SVM technically has a marginal higher value of R2 than Random Forest, it will be still safe to select Random Forest (Log2) as it has much higher reliability and lower risk of failing on new data.

So **my choice for final model is RandomForest** for the given sample of data

ALL Model trials and observations are as given below:

1.MulTIPLE LINEAR REGRESSION (R^2 **value**) = **0.7100923**

2.a. SUPPORT VECTOR MACHINE:

No Preprocessing run as default (kernel=rbf and C=1) (R^2 **value**) =-0.088427

2.b. SVM With Preprocessing

S.no	HYPER PARAMETER C	Linear (R^2 value)	RBF(NON LINEAR) (R^2 value)	POLY (R^2 value)	Sigmond (R^2 value)	PreComputed (R^2 value)
1	1	-0.010102	-0.083382	-0.075699	-0.075429	Since **pre-computed** needs a square matrix and since the one in hand is non-square 1338 X 6, we have to rule out this option.
2	10	0.462468	-0.032273	0.0387162	0.039307	
3	100	0.628879	0.320031	0.617956	0.527610	
	500	0.763105	0.664298	0.82636	0.444606	
4	1000	0.764931	0.810206	0.856648	0.287470	
5	2000	0.744041	0.854776	0.860557	-0.593950	
6	3000	0.741423	0.866339	0.859893	-2.1244	
7	10000	0.741423	0.877995	0.859171	-34.1515	
8	20000	0.741423	0.878024	0.857838	-112.4644	
9	100000	0.741418	0.872498	0.857788	-3465.95	

Observation:

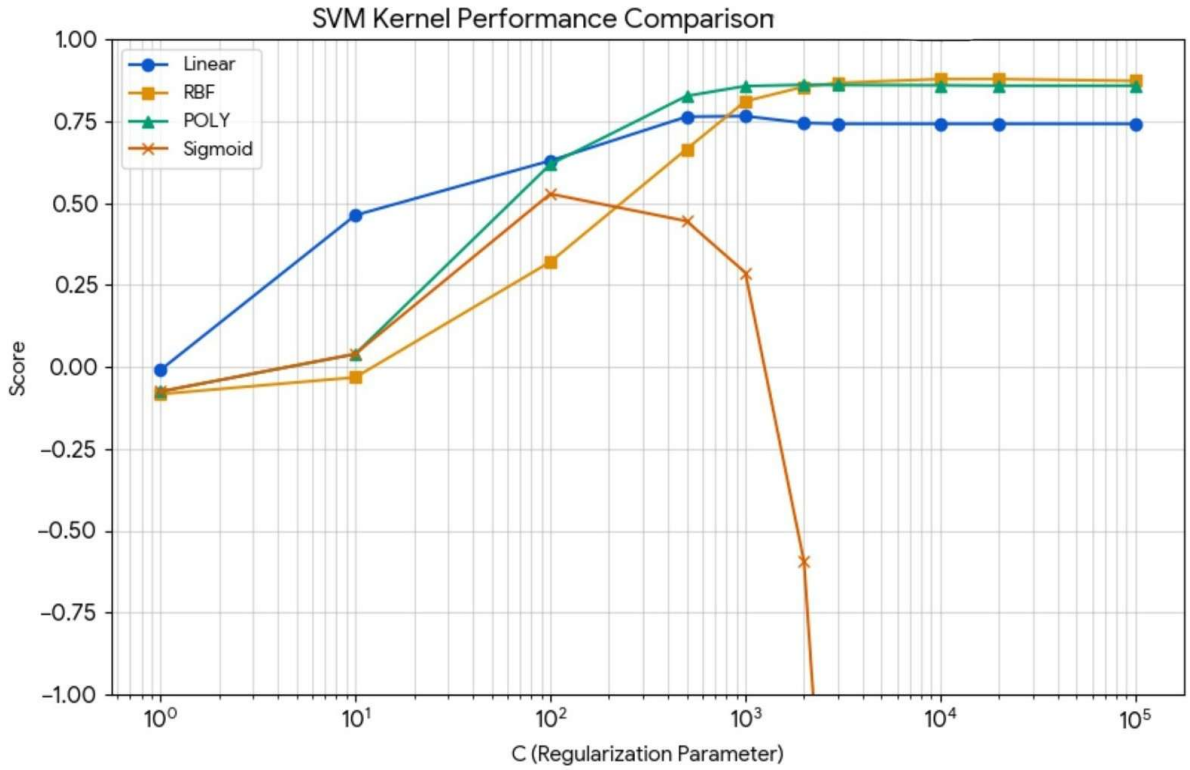
1) Kernel=Linea @ C=3000 R^2 value peaks to 74% after that it went flat till C=20,000 then it showns downward trend

2) kernel=rbf @c=20000 it peaked at 87.8% after that it is showing a downward trend

3)kernel=poly @C=2000 it peaked at 86% after that it shows a downward trend.

4)Kernel=sigmond @C=100 it peaked at 52% then it shows drastic downward trend value.

SVM Summary: None of the kernel models scored above 90% most likely we have to not use SVM Regression model for this dataset.


SVM Kernel Performance Comparison
Linear
RBF
POLY
Sigmoid
Score
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
10^0
10^1
10^2
10^3
10^4
10^5
C (Regularization Parameter)

3. Decision Tree:

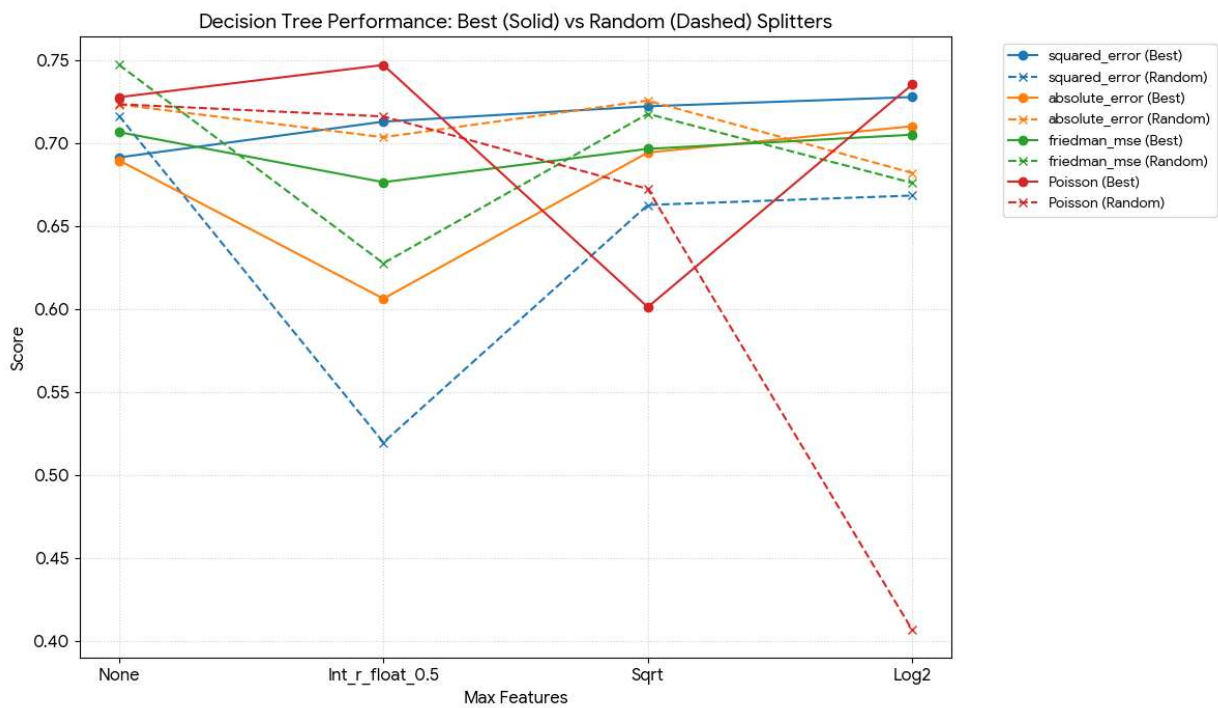
Splitter=best

S.no	MAX_FEATURES	squared_error	*absolute_error*	friedman_mse	Poisson
1	**None**	0.691525	0.689735	0.706770	0.727838
2	**Int_r_float_0.5**	0.713103	0.6064379	0.676606	0.747229
3	**Sqrt**	0.722343	0.694469	0.696677	0.601335
4	**Log2**	0.727902	0.710327	0.7052114	0.735398

Splitter=random

S.no	MAX_FEATURES	squared_error	*absolute_error*	friedman_mse	Poisson
1	**None**	0.716567	0.723431	0.747328	0.723582
2	**Int_r_float_0.5**	0.519775	0.703692	0.627757	0.716210
3	**Sqrt**	0.662881	0.725779	0.717792	0.672565
4	**Log2**	0.668550	0.682194	0.676196	0.406870

3. Decision Tree: illustration of R2-value across diff criteria



The Decision Tree with best R^2 value:

CRITERION=poisson:MAX_FEATURES=None:Splitter=random

R^2 value: 0.747328

4. RandomForest:

N_estimators=50 random_state=None

S.no	MAX_FEATURES	squared_error	*absolute_error*	friedman_mse	Poisson
1	None	0.854215	0.854000	0.846562	0.856720
2	Int_r_float_0.5	0.866949	0.870439	0.870542	0.865044
3	Sqrt	0.868622	0.869983	0.870936	0.870568
4	Log2	0.870070	**0.875757**	0.871342	0.871040

N_estimators=50 random_state=42

S.no	MAX_FEATURES	squared_error	*absolute_error*	friedman_mse	Poisson
1	None	0.856983	0.850356	0.856063	0.858135
2	Int_r_float_0.5	0.872782	0.870936	0.872112	0.871040
3	Sqrt	0.872782	0.870936	0.872112	0.871040
4	Log2	0.872782	0.870936	0.872112	0.869142

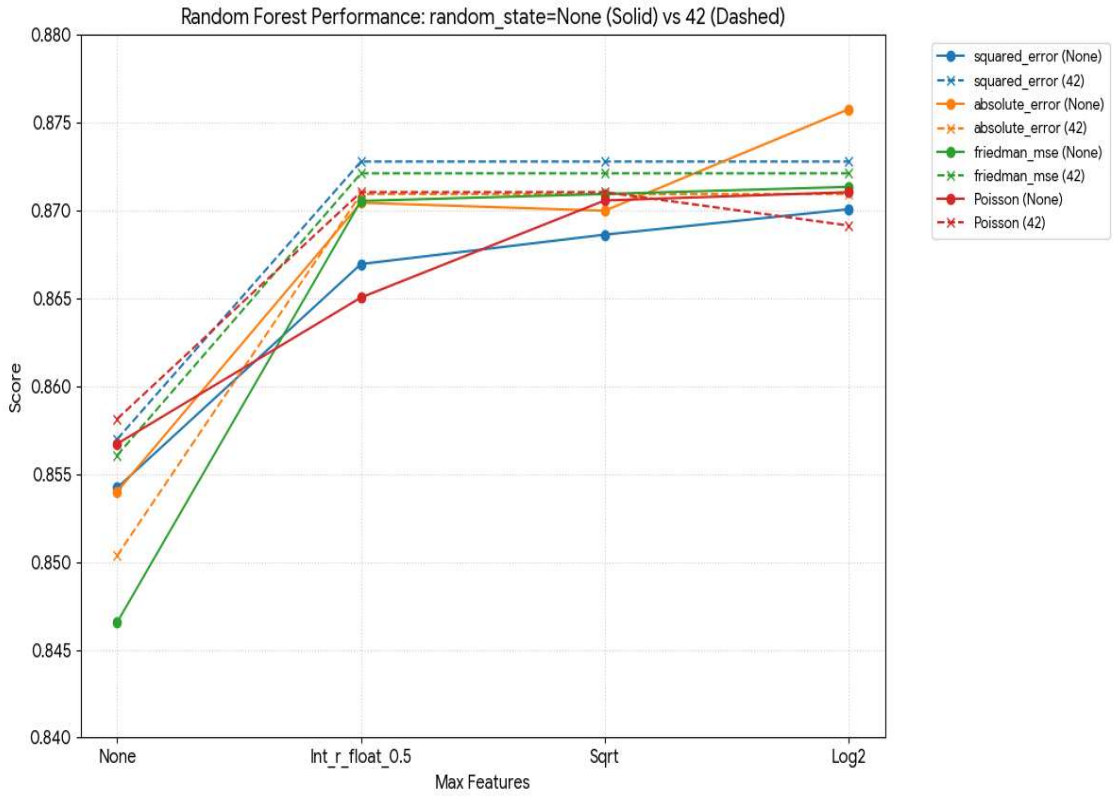
Important Observation:

When random_state is fixed at 42
for a given CRITERION(say absolute_error) the R2 value generated for all the
MAX_FEATURES(int/float,sqrt,log2) gave the same results ;
Except for MAX_FEATURE NONE

The RandomForest with best R^2 value:

CRITERION=absolute_error MAX_FEATURES=None **Random_State=None** has

0.875757 **R^2 value**


Random Forest Performance: random_state=None (Solid) vs 42 (Dashed)
0.880
0.875
0.870
0.865
0.860
0.855
0.850
0.845
0.840
Score
None
Int_r_float_0.5
Sqrt
Log2
Max Features
squared_error (None)
squared_error (42)
absolute_error (None)
absolute_error (42)
friedman_mse (None)
friedman_mse (42)
Poisson (None)
Poisson (42)